```
CREATE TABLE courses
(course_id NUMBER(10),
course_name VARCHAR2(40),
PRIMARY KEY (course_id));

 INSERT INTO courses
VALUES(101,'Data_Structures');
INSERT INTO courses
VALUES(102,'Web_Development');
INSERT INTO courses
VALUES(103,'Machine_Learning');

CREATE TABLE students
(student_id NUMBER(10),
student_name VARCHAR2(40),
course_id NUMBER(10),
PRIMARY KEY (student_id),
FOREIGN KEY (course_id) REFERENCES courses(course_id)
);

INSERT INTO students
VALUES(2,'Emma',102);
INSERT INTO students
VALUES(3,'Noah',103);
INSERT INTO students
VALUES(4,'Olivia',NULL);
```

Autocommit Display 10

```
SELECT Students.Student_Name, Courses.Course_Name
FROM Students
INNER JOIN Courses ON Students.Course_ID = Courses.Course_ID;
```

STUDENT_NAME	COURSE_NAME
Liam	Data_Structures
Emma	Web_Development
Noah	Machine_Learning

Autocommit Display 10

```
SELECT Students.Student_Name, Courses.Course_Name
FROM Students
LEFT JOIN Courses ON Students.Course_ID = Courses.Course_ID;
```

STUDENT_NAME	COURSE_NAME
Liam	Data_Structures
Emma	Web_Development
Noah	Machine_Learning
Olivia	-

4 rows returned in 0.00 seconds CSV Export

Autocommit Display 10

```
SELECT Students.Student_Name, Courses.Course_Name
FROM Students
RIGHT JOIN Courses ON Students.Course_ID = Courses.Course_ID;
```

STUDENT_NAME	COURSE_NAME
Liam	Data_Structures
Emma	Web_Development
Noah	Machine_Learning
-	Cyber_security

4 rows returned in 0.00 seconds CSV Export

Autocommit Display 10

```
SELECT Students.Student_Name, Courses.Course_Name
FROM Students
FULL OUTER JOIN Courses ON Students.Course_ID = Courses.Course_ID;
```

STUDENT_NAME	COURSE_NAME
Liam	Data_Structures
Emma	Web_Development
Noah	Machine_Learning
Olivia	-
-	Cyber_security

5 rows returned in 0.00 seconds CSV Export

Autocommit Display 10

```
SELECT *
FROM Students
NATURAL JOIN Courses;
```

COURSE_ID	STUDENT_ID	STUDENT_NAME	COURSE_NAME
101	1	Liam	Data_Structures
102	2	Emma	Web_Development
103	3	Noah	Machine_Learning

3 rows returned in 0.00 seconds CSV Export

Autocommit Display 10

```
SELECT Students.Student_Name, Courses.Course_Name
FROM Students
INNER JOIN Courses ON Students.Course_ID != Courses.Course_ID;
```

STUDENT_NAME	COURSE_NAME
Liam	Web_Development
Liam	Machine_Learning
Liam	Cyber_security
Emma	Data_Structures
Emma	Machine_Learning
Emma	Cyber_security
Noah	Data_Structures
Noah	Web_Development
Noah	Cyber_security